

RAYNEX

ONE SUN. FIVE JOBS. PAID TO SAVE WATER.

A farmer-owned solar hub for 20 smallholders on 8 ha, piloted across three farm zones of Nashik. One Nashik. One Schneider-run array keeps the cold room safe, waters by the soil, freezes ice, dries onion dries onion and sells the rest; drones and robots join as a shared service. The DISCOM shares the shares the power cost it saves when the water stays in the ground.

💧 **1,960 m³**

Water Left In The Ground
per farmer a year, -35%

🍅 **752 kg**

Harvest Saved From Rot
per farmer a year, loss cut by a third third

📈 **₹32,460**

Kept Every Year
per farmer while the loan runs,
₹55,077 after

🕒 **Payback 2.5 Years**

☀️ **72% Sun Used On Site**

✅ **₹0 Farmer Capital**

📍 **Pilot: Nashik, 3 Zones**

FIVE JOBS, ONE ARRAY, IN PRIORITY ORDER

❄️ **1 Cold Room**

💧 **2 Drip Irrigation**

📦 **3 Ice Store**

☀️ **4 Dryer**

🔌 **5 Grid Export**

\$ **stack** ATV320 Solar · Modicon M262 · LoRaWAN · Sarvam voice, 22 languages · drone + robot hire

MEET THE TEAM



Abhinav Suresh



Arpan Das



Prince Panja



Shubham Mali

IIM INDORE

Indian Institute of Management Indore

A farmer, a woman farmer in a drought taluka, a local technician and a DISCOM engineer: each wants something else from the same machine.

Ramesh Shinde, 41

Farmer, Niphad, Nashik

0.4 ha · black soil · onion, tomato

Today

☾

Power is free, so he floods at 2 am when the the feeder comes

💧

Pumps 5,600 m³ a year in a critical block, 97% 97% extracted

🕒

Loses about 1.9 t; sells at Lasalgaon on harvest day

"Water my field without a 2 am trip and sell on a sell on a good day."

Wants: sleep, +₹32,460 a year

Sunita Gaikwad, 36

Woman farmer, Yeola, Nashik

shallow soil · onion · SHG member

Today

⚠️

Waters alone at night while her husband migrates for work

🗣️

Speaks Ahirani, cannot read an app or an SMS

🏠

Onion rots in a shed; she sells to the first trader trader

"Water by day, store my onion, tell me in my my own tongue."

Wants: safety, voice, storage

Pooja Pawar, 22

ITI electrician, Igatpuri

smartphone · Marathi, English

Today

🚗

Only local option: a factory shift on the highway

🕒

Nearby solar pumps break and sit idle for weeks

📱

Wants live alerts, not a call from an angry farmer farmer

"A skilled job near home that pays every month. month."

Wants: ₹15,000 a month, a career

Er. Anil Deshmukh, 38

Assistant Engineer, MSIEDCL

Niphad sub-division · feeders

Today

📄

The state pays ₹14,761 cr a year for free farm power

⚡

Farm load peaks at night on weak rural feeders

👁️

No proof of how much water that power buys buys

"Cut the farm power bill without a farmer revolt. revolt."

Wants: a smaller bill, proof

E1: Problem Statement · Personas

Contrasting Needs, One Hub: What Each Feature Gives Each User

Exhibit F6, Slide 11

Feature	Ramesh, Farmer	Sunita, Woman Farmer	Pooja, Technician	Anil, DISCOM
Daytime Solar Drip	No 2 am trips; −35% water	Waters by day, near home	Fewer night fault calls	Night load moves to day
Cold Room, Sell-Or-Store	Holds tomato for + ₹3/kg	Onion stored, not dumped	Watches room temperature	Cold load runs on own solar
Voice In 22 Languages	Marathi call at 7 am	Ahirani voice, no reading	Marathi and English app	English portal
Bachat Bonus	Share of ₹46,968 via FPO	Same share, own vote	Meters she keeps prove it	Pays only on metered saving
Plot Score And Alerts	One spoken action a day	Drone Didi sprays weak patches only	Alert to her phone in 15 min	Feeder data without site visits
FPO Owns The Hub	₹0 capital	One member, one vote	Paid ₹15,000 a month	One counterparty, 20 farmers

Design Rules From Four Users

🗣️

Voice Before Screens

Ahirani and Marathi first

☀️

Daylight Before Night

No pump runs after dark

📊

Proof Before Payment

Meters, not trust

👤

Owners Before Automation

Recommend mode, one vote

Water, sun and harvest all leak for the same reason: the next unit costs the farmer nothing and saves him nothing.

Free marginal water, idle solar and no farm-gate cold cost India ₹2.8 lakh crore a year.

E1: Problem Statement · Root Cause

Leak 1: Water Nobody Pays For

Groundwater extracted, % of recharge, Nashik Nashik

Sinnar	99.9%
Niphad	97.4%
Deola	95.0%
Baglan	83.6%
Yeola	79.0%
District	58.5%

Maharashtra pays ₹14,761 cr a year so 44.06 lakh pumps run free

Next hour of pumping costs the farmer ₹0; the DISCOM pays ₹5.54 a unit

States spent ₹1.89 lakh cr on power subsidy in FY25 (CAG)

India extracts 245.6 BCM a year; 11% of blocks over-exploited

→ Exhibit F2 Subsidy Graph, Slide 11

Leak 2: Solar That Sleeps

Standalone solar pump, days a year

150 pumping	215 idle
-------------	----------

35%

Solar used on site, Dhundi pilot

40%

Gujarat SKY farmers who pumped less

Solar fixes the fuel bill but makes the next hour of pumping cheaper still

SKY paid for surplus power; selling water paid ₹6 to ₹8 a unit, so 60% kept pumping

No battery the farmer can afford pump-sized lithium wrecks payback

215 idle days: the asset earns nothing for 59% of the year

→ Exhibit F4 Lessons Abroad, Slide 11

Leak 3: Harvest That Rots

Loss from field to market, % of output

Fruit, max	15.1%
Tomato	11.6%
Onion	7.3%
Cereals	5.9%

Tomato is the worst vegetable; loss across crops ₹1.53 lakh crore a year

Nashik grew 58.8 lakh t of onion in 2024-25; Maharashtra is 35.5% of India

No farm-gate cold, so Ramesh sells at Lasalgaon on harvest day

What India loses a year, ₹ crore

Power subsidy	1,30,000
Harvest loss	1,53,000

→ Exhibit A5 Loss Baseline, Slide 12

First Principles: Six Questions

Why over-pump? Power is free up to 7.5 HP, so the next hour costs him nothing

Who loses most? The state, ₹1.3 lakh crore a year, so the state should pay for water saved

Why do tomatoes rot? No farm-gate cold; the cheapest battery is a tank and ice

Why no adoption? Cost, trust and reach: share the asset, speak the advice

Why is efficiency not enough? Saved water gets spent on more land; cap the total, not water per ha

Who should own it? The farmers, one member one vote: Denmark had 1,163 co-op dairies by 1909

Root cause: free marginal water, and saving it pays no one.

Climate Makes It Worse, Rails Make It Fixable

Monsoon 14 to 15% short in 2026, with more heat

Farms: 40%+ of jobs, ~90% of freshwater

Schneider: 15 to 20% of produce value lost

Nashik alone spans 550 to 3,400 mm of rain

Digital rails already built, crore

Crop plots surveyed	31.3
Farmer IDs	10.31
SMS monsoon alerts	3.88 (31 to 52% acted)
Kisan e-Mitra queries	0.93 in 11 languages

One farmer-owned array, a Schneider router that sends every unit of sun to the job worth most, and a bonus for water saved.

Nine parts, one FPO owner, one DISCOM contract: every Schneider goal gets a number.

E2: Solution · E3: Alignment

The Hub, In Priority OrderExhibit A3, Slide 12



 1 Cold Room

5 t room guards up to 250 crates; 2,700 kWh

 2 Irrigation

Soil-gated drip, 8 zones; 15,430 kWh kWh

 3 Ice Store

Midday ice keeps the room cold at night

 4 Dryer

Onion and chilli; 300 kWh

 5 Grid Export

7,070 kWh at ₹3.05, plus the bonus

72% of the sun used on site, against 35% for a solar pump alone

Nine Components, One Contract

 C1

Energy Router

15 kWp, ATV320 drive, picks the load

 C2

Raised Array

3.5 m stilts over the nursery, ≤30% shade

 C3

Soil-Gated Drip

–35% water, +20% yield, 50 m³ tank

 C4

Bachat Bonus

DISCOM pays 50% of the subsidy saved

 C5

Cold-Chain Ladder

₹6,000 chambers, then a 5-tonne room

 C6

Plot Score

One 0 to 100 score per plot, daily

 C7

Drone Didi

SHG pilots spray only flagged patches

 C8

Voice First

Voice in 22 languages, from Marathi

 C9

FPO Operator

20 farmers, 8 ha, one member one vote

New: C1 router chooses the load itself, and C4 pays for water not pumped.

→ Exhibit F5 Blue Ocean, Slide 11

Schneider's Four Goals, Each With A Number

1

Cut Energy And Water Intensity

–35% water; 3.57 → 6.59 kg/m³; 100% solar

2

Minimise Post-Harvest Loss

Tomato loss 11.6% → 7.7%; 15.04 t saved per hub

3

Empower The Smallholder

₹0 capital, voice in Marathi, +₹32,460 a year

4

Strengthen Climate Resilience

Tank, ice, AquaCrop; 39,200 m³ a hub kept underground

Schneider's Five Starter Ideas: All Five In

 Solar precision irrigation C1, C3

 AI crop health C6

 Solar cold chain + logistics C5

 Vernacular voice C8

 Aggregation, co-ops C9

Schneider's Condition, Answered

 Software: Energy router + Plot Score

₹ Business model: FPO-owned hub, Bachat Bonus, AIF loan

 Farmer tools: Missed call, voice, SMS, a village operator

Why Not A Simpler Fix?

Option	Sun Used On Site	Water	Harvest	Who Pays
Solar pump alone	35%, idle 215 days	Pumps more: 60% unchanged in SKY	No cooling	Farmer + subsidy
Free grid pump, today	0%	Floods at night, free	No cooling	State: ₹14,761 cr a year
Cold room alone	Grid power	No change	Loss cut	Farmer
RAYNEX hub	72%, five loads	–35%, capped by budget	Loss cut by a third	FPO, AIF loan, DISCOM bonus

Three design artifacts: a service blueprint for the season, the missed-call voice flow, and four offline app screens.

No smartphone needed: one missed call, one answer in their own language, one action a day.

E4: User Journey · D3: Design Artifacts

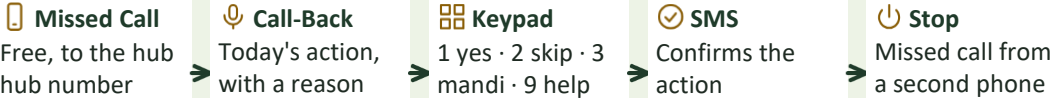
Service Blueprint: Five Lanes Through One Season

Exhibit F6 JTBD And Rogers, Slide 11

Lane	Before Sowing	Growing Season	Harvest	Off-Season
Farmer	Joins the FPO with his Farmer ID; voice onboarding	Morning voice call; approves or overrides (recommend mode)	Drops crates at the hub; hears sell-or-store	Input-cost alerts; books seed and fertiliser in bulk
Operator	90-second soil scan; installs probes at 15 and 45 cm	Clears alerts within 24 h; Pooja backs 5 hubs	Weighs every lot in and out on the platform scale	Cleans panels; services the cold room
Router, Edge	Loads the crop plan and the season water budget	Waters to the soil balance; caps pumping at budget	Cold room first, then ice store	Dryer runs, surplus exported
Cloud, Data	AquaCrop plan; NDVI zones from Sentinel-2	Plot Score 0 to 100, daily, per plot	Shelf life per lot; route and truck pooling	Season report and metered bonus proof
Partners	AIF bank loan; KVK Nashik	Drone Didi spray bookings	Lasalgaon buyers, eNAM	DISCOM pays the bonus; insurer gets consented data

Missed-Call Voice Flow: Marathi, Ahirani, Hindi At Launch

"Rameshram, aaj zone 2 la 40 minit paani dya. Udaya paavsaachi shakyaata 70 takke." takke." (Marathi: water zone 2 for 40 minutes; 70% chance of rain.)



Four App Screens: Flutter, Offline-First, Icon-Led (illustrative)

1 Today

Z1

Z2

Z3

Z4

Pump: solar on

Rain tomorrow 70%

Do: zone 2, 40 min

Press 1: Approve

2 Store Or Sell

Lot 14: 4 days left

Today ₹10/kg

Hold: +₹3/kg

Do: store, sell Friday

Store This Lot

3 Earnings, Budget

Sun: made, used, sold

Water vs budget

Bonus: on track

See This Season

4 Crop Health

Plot Score 78

NDVI: 2 weak patches

Leaf photo check

Do: book Drone Didi

Book Drone Didi

Ramesh's Day, Then
And Now
Zero-night trips

2 AM

Wakes for feeder power,
floods the field

Asleep: tank and ice did the
night shift

0 Night Trips

7 AM

Guesses what to water

Voice call: zone 2, 40
minutes

1 Action A Day

11 AM

Sun on an idle pump

Router runs drip on solar, by
the soil

−35% Water

2 PM

Picks tomatoes in the heat

Crates into the cold room

250 Crates Safe

5 PM

Sells at today's price

Sell-or-store says hold

+₹3/Kg On Stored Lots

9 PM

Tomatoes start to soften

Room on ice; ledger updated

Loss Cut By A Third

A Schneider ATV320 drive and a Modicon M262 router run every load; data decides on the farm; money loops via FPO, bank, MSEDCL.

Energy goes where it is worth most, data decides on the farm, and money pays for water left in the ground.

Where The Sun Goes, kWh A Year

25,500 kWh

Irrigation

15,430 · 60%

Cold Room

2,700 · 11%

Dryer

300 · 1%

Grid Export

7,070 · 28%

72% used on site, against about 35% for a solar pump at Dhundi

Fail-Safe By Design

✓ Edge controller runs pump, cold band and budget cap **on the farm**

⚡ A network drop **never** stops water or cooling

🔌 Manual override on pump and cold room

👤 **Pooja**, Hub Mitra for 5 hubs, fixes alerts within 24 h

System Architecture: Five Layers, Three Flows

Energy

Data

Money

Exhibit A3, Slide 12

L5
People

Farmers: Voice, 22 Languages

1 Action A Day

L4
Intelligence

FAO-56, AquaCrop, NDVI

Daily, Per Plot

L3
Post-Harvest

S4S Dryer, Chambers

300 kWh Dryer

L2
Field

Drip, 8 Zones, Fertigation

-35% Water

L1
Energy

15 kWp TOPCon DCR, Stilts

25,500 kWh/yr

FPO: Owns Hub, 20 Farmers

Plot Score 0 To 100

Ecofrost 5 t, Thermal

10 HP AC Pump, 50 m³

ATV320 Solar, 7.5 kW

AIF Bank: ₹4.52 L A Year

Shelf Life, Sell Or Store

250 BLE Crate Tags

LoRa Probes, Flow, Piezo

Modicon M262 Router

MSEDCL Portal, Insurer

AWS Mumbai, 4G Sync

UG65 Gateway + Edge

EM6400NG + Net Meter

₹46,968 Bonus

Per Lot, Per Day

4G Sync

Every 15 Min

Priority 1 To 5

Proof

4G Sync

7,070 kWh Out

Built On Schneider Parts

ATV320 Solar: MPPT, Dry-Run Guard, To 50°C

SMS On/Off + EcoStruxure Livelihood

Modicon M262 Runs The Router

EM6400NG Meters: Bonus Proof

Up To 30% Less Pump Energy

SCHNEIDER INSIDE

Router Priority, Every Few Minutes

❄️ 1 Cold Room: ₹53 of power guards ₹50,000 of crop

💧 2 Irrigation: zones flagged dry

📅 3 Ice Store: midday ice for night

☀️ 4 Dryer: onion and chilli

🔌 5 Grid Export: at ₹3.05 a kWh

Money Loop, Per Hub A Year

₹ Farmer Fee ₹30,117 → FPO

🏠 FPO → Bank ₹4.52 L AIF

📄 MSEDCL → FPO ₹46,968 Bonus

🏛️ MSEDCL → FPO ₹21,564 Export

🏧 Subsidies ₹9.10 L at 80%

Ramesh's Ledger every year

👤 **Ramesh, 41**
0.4 ha in Niphad, Nashik

💧 1,960 m³
water left in the ground

🍅 752 kg
tomato and onion saved

₹ ₹32,460
kept a year during the loan

🌙 0
night trips to the pump

Off-the-shelf hardware, open-source software, and a voice stack that speaks 22 Indian languages by Tier 3.

🕒 D4: Prototype · E5: Technical Approach

Hardware Bill Of Materials			Exhibit A7, Slide 12
Component	Model And Spec	₹ Indicative	
PV, 15 kWp	26 × 580 Wp TOPCon, DCR	3.6 to 4.1 L	
Stilts, 3.5 m	HDG steel, 80 µm coat	2.1 to 2.7 L	
Pump	10 HP AC submersible	3.5 to 4.5 L	
Solar Drive	Schneider ATV320 Solar	50 to 75k	
Router PLC	Schneider Modicon M262	25 to 60k	
Meters × 3	Schneider EM6400NG	7 to 13k ea	
Soil Nodes × 8	LoRaWAN moisture + EC	8 to 15k ea	
Flow Meter	Kritsnam Dhaara, ultrasonic	31 to 38k	
Borewell Level	Piezometer, telemetry	18.5k	
Gateway, Edge	Milesight UG65 + 4G router	60 to 85k	
Weather Station	Rain, heat, RH, wind, sun	0.5 to 1.5 L	
Cold Room, 5 t	Ecozen Ecofrost, 30 h backup	11.4 to 15 L	
Dryer	S4S conduction, 100 kg	3.6 L	
Drip, 8 ha	Jain or Netafim, fertigation	8.96 L	
Gross ₹37.16 L a hub; gold rows are Schneider parts. Tenders now buy a solar pump at pump at about ₹2.4 L.			

Software Stack, Top To Bottom	
APPS	Flutter app, offline, under 42 MB · IVR · WhatsApp WhatsApp · FPO dashboard · MSEDCL portal portal
AI, VOICE	Sarvam speech in and out · Sarvam-105B on ICAR advice · Plot Score model
BACKEND	Python FastAPI · TimescaleDB · AquaCrop-OSPy + OSPy + FAO-56 · Sentinel-2, IMD, Agmarknet, Agmarknet, AgriStack APIs
CLOUD, IoT	AWS Mumbai, MeitY-empanelled · AWS IoT Core, \$1 per million messages · ChirpStack LoRaWAN · ThingsBoard
EDGE	Python and Node-RED on the gateway · MQTT store-and-forward · pump, cold band and budget cap run locally
DEVICE	LoRa soil nodes · Modbus meters · ATV320 · M262 · BLE crate tags
🔒 DPDP Act 2023: farmer-owned data, consent before any any insurer or lender sees it	
📶 A network drop never stops water or cooling	

Voice In 22 Languages		Exhibit A10, Slide 12
Speech In Sarvam Saaras v3 22 languages, phone phone audio, ₹30 an an hour	Reasoning Sarvam-105B 22 languages, Apache 2.0, grounded on ICAR	Speech Out Bulbul v3, Parler- TTS 11 + 21 languages, languages, ₹30 per per 10K chars
Tier 1, Launch	Marathi, Hindi, Ahirani (fine-tuned on 5 to 7 h of speech)	
Tier 2, Month 4 Month 4 To 9 9	Telugu, Kannada, Gujarati, Bengali, Tamil, Odia, Punjabi, Malayalam; Bhili, Bundeli, Bhojpuri, Malvani	
Tier 3, All 22	IndicConformer + IndicTrans2 + Parler-TTS; TTS; Bhashini as failover	
₹	About ₹7 to 8 per 3-minute call; cached answers cut the costliest part, speech out	
👤	Demand is proven: Kisan e-Mitra answers 8,000+ queries a day in 11 languages	
📞	Low speech confidence? Keypad menu, then an agronomist calls back	
⚠️	Bharat-VISTAAR speaks only Hindi and English so far; BharatGen Param2 is non-commercial	

D4 Software
Prototype By 22 Nov
Hand-coded, no hardware

⚡ 1 Router Simulation
Hourly Nashik sun for a year; 72% vs 35% on site

out: Monthly kWh by load

💧 2 Irrigation Scheduler
FAO-56 + probes + rain-skip; AquaCrop-OSPy runs

out: Minutes per zone, daily

🗣️ 3 Voice Demo
Marathi and Ahirani missed-call IVR on a test number

out: One spoken action + SMS

🎯 4 Plot Score Demo
Sentinel-2 NDVI for one Niphad village + mock probes

out: Score and weak patches

🏠 5 Sell-Or-Store
Onion and tomato storage + Lasalgaon prices

out: Hold or sell, best market

No smallholder can own a drone or a robot; an FPO hub can, and rents them out by the acre, charged on the same solar array.

China flies 251,000 farm drones; India's Drone Didi scheme has delivered 1,094. A shared hub closes that gap.

E6: Innovation · Automation Layer

Drones: Live From Year 1

Exhibit A9, Slide 12



Metric	Manual Knapsack	Drone, 10 to 16 L
Cost per acre	₹800 to 1,500	₹400 to 800
Water per acre	200 to 250 L	10 to 15 L, −94%
Speed	a day per acre	1 acre in 15 min

- ☉ Sprays only the weak patches the **Plot Score** flags; scouts with a with a multispectral camera
- ⚡ Batteries charge on hub solar at midday, when the router has surplus
- ₹ **Funding:** Namo Drone Didi **80%** up to **₹8 L**; SMAM **75%** for FPOs FPOs
- 👤 Pilot earns **₹400 to 600** an acre; Igatpuri hub also hauls produce produce downhill

Robots: Phased In From Year 2



Y2 Spot-Spray Robot
Camera finds each weed or sick plant
Niqa: **1.48 lakh** acres in 2024-25, Akola, MH;
MH; inputs −60%



Y3 Grading Line, Robot Arm
Sorts onion by size and rot into crates
Feeds shelf life per lot; cuts packing labour



Y5 Autonomous Weeding
US See & Spray: **5 mn+** acres, ~50% less herbicide
Hired, never owned: the FPO runs a custom-hiring centre

- 🔄 Plot Score flags a patch → app books a machine → it charges on hub solar → works → proof logged
- 🏠 AIF funds custom-hiring centres and drones; machines are AIF-eligible assets

India Vs The World

Exhibit F4, Slide 11

Farm mechanisation, %

USA	95%
Brazil	75%
China	60%
India	47%

Farm drones in use, log scale

China	2,51,000
India	1,094 Drone Didi

Netherlands Greenhouse tomato **500+ t/ha** vs India ~25

Israel **90%** micro-irrigated vs India **5.5%**; flying fruit pickers

Japan Drone-sprayed area up **175x** in 5 years; farmers average **67.8**

USA FAA lets **3-drone** swarms spray **150** acres an hour

⚠️ India: **46%** of workers, **16%** of GDP; Drone Didi at **7%** of its **15,000** target

Automation Roadmap

Machines rented; land stays the farmer's

Y1 🛩️ 1 Drone, 1 Pilot

Spray and scout; Sunita's SHG flies it

₹8 L subsidy cap

Y2 ☉ Spot-Spray Robot

Hired per acre from a cluster of hubs

inputs −60%

Y3 🏠 Grading Line

Onion and tomato sorted at the cold room

shelf life per lot

Y4 🚚 Autonomous Cart

Moves crates field to cold room

labour −30 to 40%

Y5 ☀️ Weeding Robot

Rented across 25+ hubs

herbicide ~−50%

Pay To Save Water, And What It Returns

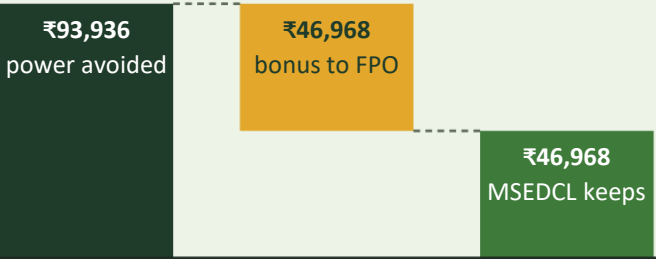
Baseline: a Niphad smallholder, tomato in kharif and onion in rabi, flood irrigation on a free-power grid pump, no farm-gate cooling.

A metered bonus makes saving the best move; a hectare returns ₹1,37,693 a year on 35% less water.

✓ E6 Innovation · D5 Benefit · E7 Impact

What Is New: The Bachat Bonus

Per hub: 16,956 kWh × ₹5.54 power purchase cost



- 1 Borewell sensor + flow meter set the season's m³ budget budget
- 2 Controller caps pumping at budget: no rebound
- ₹ 3 MSEDCL pays 50% of the saving; booked at ₹37,574

Gujarat SKY	Paid for power; 60% kept pumping
Punjab PBPK	₹4/kWh saved; 4.69% enrolled
RAYNEX	Metered, automatic, default-on
⚠	Bonus is only 3.4% of base benefit: payback 2.6 yrs without it

→ Exhibit F1 Payoff Matrix, Slide 11

Per Hectare Per Year, Against Baseline

Metric	Baseline	RAYNEX	Change
Water Pumped	14,000 m³	9,100 m³	−35%
Free Grid Power	2,120 kWh	0, solar	−2,120 kWh
Tomato Water Use	3.57 kg / m³	6.59 kg / m³	+ 85%
Yield, Tomato / Onion	25 / 18 t	30 / 21.6 t	+ 20%
Loss, Tomato / Onion	11.6% / 10%	7.7% / 6.7%	1.88 t saved
Energy Per Quintal	4.93 kWh	3.74 kWh	−24%
Grid Export	0	884 kWh	+ 884
CO₂ Avoided	n/a	2.14 t	2.14 t
Net Benefit	n/a	₹1,37,693	₹81,150 loan yrs

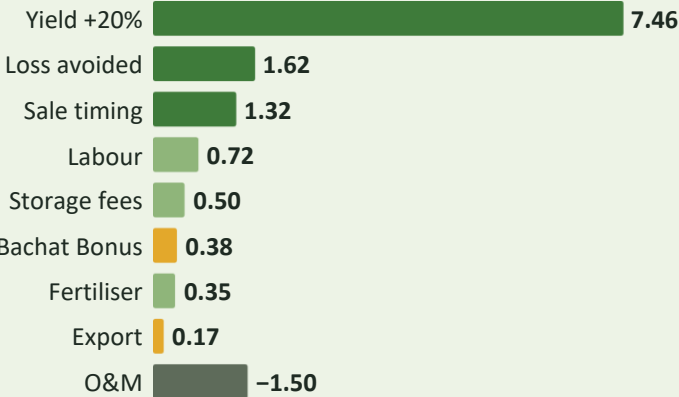
- 💧 Drip needs 10 m more head, so 35% less water saves 9% of pump energy; the sun supplies the rest
- ↩ Conservative (+10% yield, no bonus, no sale timing): ₹54,810 a hectare

→ Exhibit A1 And A4, Slide 12

Where The Money Comes From

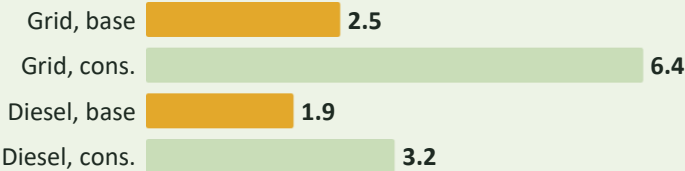
Exhibit A7, Slide 12

Per hub of 8 ha, ₹ lakh a year



Net, base ₹11.02 L	Net, conservative ₹4.38 L
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Payback on ₹28.06 L net capex, years



Per Hub → At 500 Hubs

💧 Water 39,200 → 1.96 cr m³	🌿 CO₂ 17.1 t → 8,529 t	🍅 Food 15.04 t → 7,520 t	👥 Jobs 1.2 → 600	🏠 MSEDCL saves ₹4.70 cr
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Every Player Gains
a year, per hub

👤 Farmer +₹32,460, then +₹55,077	👥 FPO Fee book ₹6.02 L	🏢 MSEDCL Net +₹46,968	🏦 Bank ₹25.25 L, Gol pays 3%	🌞 Rural Youth ₹15,000 a month	⚡ Schneider ₹1.2 L a hub	💧 Aquifer 39,200 m³ kept
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One district with three farm zones proves the hub works on every soil; existing schemes and a farm loan pay for it; partners each bring one part.

3 hubs in 3 Nashik talukas from April 2027, ₹0 farmer capital, 500 hubs by year 5.

D6: Deployment And Scale-Up · E8: Roadmap

Why Nashik: Three Zones, One District

Exhibit A8, Slide 12

📍 Hub 1: Igatpuri Western Ghat zone Red, lateritic soil · very high rain rain Groundwater safe Paddy on AWD + vegetables Tests: wet, hilly hub	📍 Hub 2: Niphad Plains, irrigated Medium-deep black soil Groundwater 97.4%, critical Onion, tomato, grape · Lasalgaon Tests: anchor grid hub	📍 Hub 3: Yeola Scarcity zone Shallow soil · low rain Groundwater 79.0%, semi-critical critical Onion, rainfed crops Tests: drought hub, farm pond
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- 📖 Nashik spans 3 of Maharashtra's 9 agro-climatic zones and 550 to 3,400 mm of rain: a "mini-Maharashtra"
- 🕒 58.8 lakh t of onion a year and Lasalgaon, Asia's largest onion market; free power for 44.06 lakh state pumps

What It Costs And Who Pays

Exhibit A6, A7, Slide 12

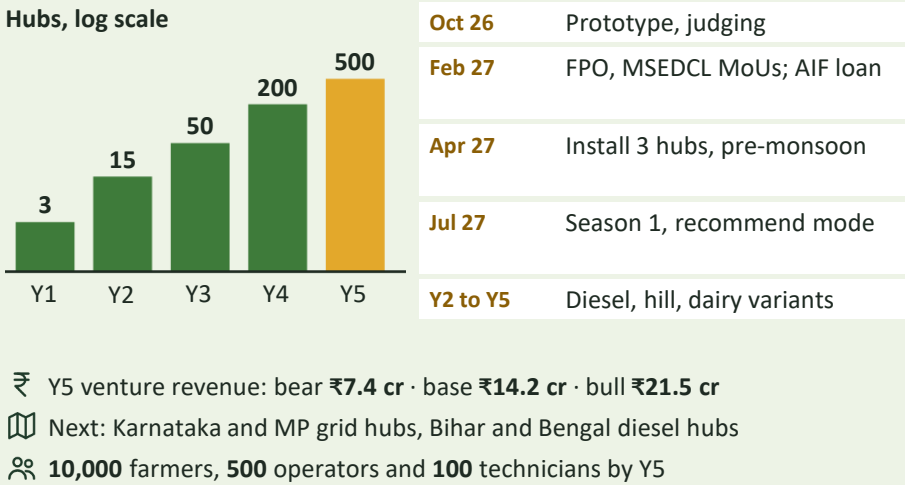
Schemes ₹9.10 L	AIF loan ₹25.25 L at 6%, 7 years
Gross capex per hub ₹37.16 L; farmer capital ₹0	
FPO grant ₹2.81 L	
☀️ PV: KUSUM 2.0, about ₹50,000 cr with 10 GW agri-PV, awaits notification; fallback Magel Tyala, 90%	
💧 Drip: Per Drop More Crop, 55% for small farmers	
❄️ Cold room: MIDH 35%; up to 75% stacked with RKVY (upside)	
🏠 Loan: AIF, 3% interest subvention and a credit guarantee	
₹ Repayment: fee ₹30,117 per farmer a year covers the ₹4.52 L instalment and O&M	
Pilot of 3 hubs: ₹1.11 Cr gross, ₹84 L net of subsidy, all booked at 80%	

Partnerships, Each With A Fallback: No Grant Asked

Partner	Brings	Gains	Fallback
Schneider Electric	ATV320, M262, meters, design support	A DISCOM product, a reference design	Ecozen Ecotron
MSEDCL, Nashik Zone	Net meter, ₹3.05 export, bonus MoU	₹46,968 net a hub	Season-1 bonus waived
Sahyadri Farms FPC	30,000 farmers, market access	Storage, member value	A Niphad FPO
KVK Nashik, MPKV Rahuri	Agronomy, field trials	Field data	NHRDF
NABARD, AIF Lender	₹25.25 L loan, guarantee	Subvented, secured loan	SFAC
Ecozen	Ecofrost cold room	Install base	Inficold
Sarvam, Bhashini	Voice in 22 languages	An agri use case	AI4Bharat models
Lasalgaon APMC, eNAM	Prices, buyers	Steadier arrivals	FPO pooled sales

Rollout And Roadmap

Exhibit F8, Slide 11

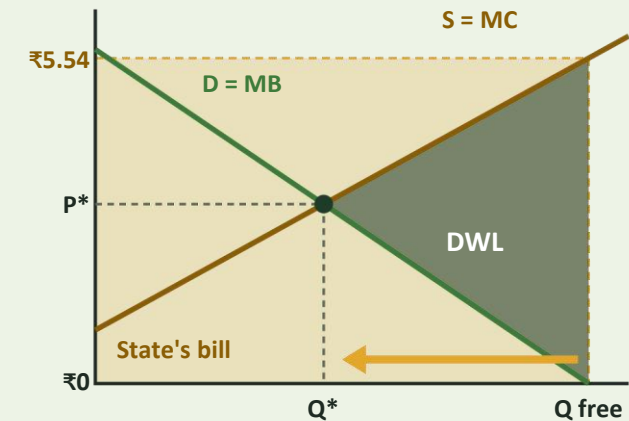


EXHIBITS F1 TO F8: THE FRAMEWORKS BEHIND THE CASE

Public policy, micro and macroeconomics, strategy and projections: why the state should pay, how much, and why the numbers hold.

F2 Microeconomics: The Subsidy Graph

Farm power: ₹/kWh vs kWh pumped



Gold arrow: the bonus pulls use back to Q*

- ⚡ **Free power:** the farmer pays ₹0, so he pumps to Q free, where the last unit is worth nothing to him
- ₹ **State's bill:** MSEDCL still pays ₹5.54 a unit; the state covers the box, ₹14,761 cr a year
- ⚠ **Deadweight loss:** every unit past Q* costs more than it is worth
- 🔧 **Our fix:** a Pigouvian subsidy on saving. Each unit not pumped earns ₹2.77, half of ₹5.54, so use moves back toward Q*

📄 **Evidence:** Punjab's ₹4 a unit cut pumping 7.5 to 30%

F3 Public Policy: How Much Should The State Pay?

Government Ledger, Per Hub	One-Time	Every Year
Capex: KUSUM 2.0, PDMC, MIDH	₹9.10 L	n/a
Power purchase avoided	n/a	+ ₹93,936
Bachat Bonus to FPO	n/a	− ₹46,968
AIF 3% subvention, GoI	n/a	− ₹75,756 Y1, falling
Net to MSEDCL	n/a	+ ₹46,968

Self-financing: the state pays for water not used out of power it no longer buys; states spent ₹1.89 lakh cr on power subsidy in FY25.

- 🔧 **Our call:** no new capex money. Bonus = 50% of the metered saving, paid after the season
- 🎯 **Bounds:** cost of saving < bonus ≤ ₹93,936; 50% splits the surplus (Nash bargain)
- 💧 **Water price:** ₹0 → a shadow price of ₹1.20 per m³ saved (₹46,968 ÷ 39,200 m³)
- 🍅 **Food prices:** stored lots sell +₹3/kg off-peak, easing Lasalgaon gluts
- ⚡ **Power price:** MSEDCL buys day solar at ₹3.05, below its ₹5.54 average

F1 Game Theory: Payoff Matrix

FPO / MSEDCL payoff, ₹ per hub a year

	Bonus	None
Save	+46,968 / +46,968	0 / +93,936
Pump	0 / 0	0 / 0, today

🎯 Without a bonus, pumping is dominant; with one, (save, pay) is the Nash equilibrium

✅ MSEDCL pays only on proof

Who moves first: FPO grant and AIF fund 3 pilots → meters prove → MSEDCL signs → AIF lends hubs 4 to 50

🔄 **Repeated game:** a payout each season builds trust

F8 Projections And Viability

Market, ₹ crore, log scale

TAM	19,600
SAM	18,000
SOM	185.8

TAM: 49 lakh KUSUM pumps · SAM: 45 lakh Maharashtra farm pumps · × ₹40,000

Hubs Y1 to Y5: 3 · 15 · 50 · 200 · 500
Farmers: 60 → 10,000

Y5 revenue: bear ₹7.4 cr · base ₹14.2 cr · bull ₹21.5 cr

🔧 **MSEDCL at 500 hubs:** avoids ₹4.70 cr, pays ₹2.35 cr a year

🕒 **Viability:** ₹11.02 L a year on ₹28.06 L net: payback 2.5 yrs

F4 Lessons From Abroad And Home

Denmark	1,163 co-op dairies by 1909: copy one member, one vote
Israel	Quotas, block rates: copy the budget
Gujarat	Paid for power only: avoid
Punjab	4.69% enrolled, late pay: make it default, automatic
China	2.51 lakh drones on 178 mn ha: share machines

F6 JTBD And Rogers Diffusion

- 🔧 **JTBD:** sleep, safety, a job, a smaller bill
- ➡ **Rogers:** FPO board first; recommend mode to try; neighbour plots to see

F5 F7 Strategy, Macro, PESTEL

Now: KUSUM 2.0 pending

1 to 5 yrs: ₹50,000 cr, 10 GW agri-PV

5 to 15 yrs: free power ends 2029

P Free farm power **E** ₹1.89 lakh cr

S 82% smartphones **T** 22-language AI

E Aquifers falling **L** DPDP, ALMM-II

🎯 **7 Powers:** counter-position, switching costs, feeder data

🔧 **Porter:** rivals Ecozen, Fasal; barrier, the DISCOM contract

⭐ **Blue Ocean:** create the bonus, eliminate night pumping

✅ **Moat:** the DISCOM contract, not the box

EXHIBITS A1 TO A10: EVERY ASSUMPTION, IN ONE PLACE

Every number in slides 2 to 10 traces to a line here; (A) marks our model, the rest are published figures. Scheme money is booked at 80%.

A1 Baseline, Per Hectare

Crops	Tomato K, onion R
Flood Water, A Season	7,000 m³
Water A Year	14,000 m³
Pump Lift (A)	25 m
Grid Power A Year	2,120 kWh
Farm-Gate Cooling	None
Yield, Tomato / Onion	25 / 18 t
Energy Per Quintal	4.93 kWh

A5 Loss And Storage

Loss, Tomato / Onion (A)	11.6% / 10%
Loss Cut (A)	One-third
Food Saved Per Hub	15.04 t
Sale Timing, Stored	₹3/kg on 44 t
Storage Fees	₹50,000
Crates Guarded	250 × 20 kg
Cold Room Draw	15 kWh a day
Onion Loss, NABCONS	7.27%

A9 Manpower, O&M, Drones

Operator, Per Hub	₹4,000 a month
Technician, Per 5 Hubs	₹15,000 a month
Platform Fee	₹60,000 a year
Spray Drone, Certified	~₹10 L
O&M Total, Per Hub	₹1.50 L a year
Drone Didi Subsidy	80%, ₹8 L cap
Team Y1, Y3, Y5	5 / 12 / 35
Labour Cost Cut, Drip	30 to 40%

A2 Farm And Hub

Farmers Per Hub	20
Area Per Hub	8 ha
Ramesh's Plot	0.4 ha
Ramesh's Revenue	₹1,86,400
Hub Crop Revenue	₹37.28 L
Tomato Price	₹10/kg
Power To The Farmer	Free to 7.5 HP
FPO Equity Grant	₹2,000 a member

A6 Policy Money, Booked At 80%

KUSUM 2.0 Or Magel, PV	30%
PDMC, Drip	55%
MIDH, Cold Room	35%
Booked Subsidy	₹9.10 L
MSEDCL Power Cost	₹5.54/kWh
Bachat Bonus	50% of ₹93,936
Bonus Booked	₹37,574
Equity Grant Cap Per FPO	₹15 L

A10 Rollout And Revenue

Hubs Y1 To Y5	3, 15, 50, 200, 500
Integration Margin	₹3.7 L a hub
Schneider Hardware	₹1.2 L a hub
Voice, Per 3-Min Call	₹7 to 8
Install Window	Apr to Jun
Pilot Talukas	Igatpuri, Niphad, Yeola
Margin Money, 10%	₹2.81 L
Y5 Revenue, Base	₹14.2 cr

A3 Energy Model (A)

Array	15 kWp
Yield Per kWp	1,700 kWh
Generation A Year	25,500 kWh
Water, Cold, Dryer kWh	15,430, 2,700, 300
Export At ₹3.05, MSKVY	7,070 kWh
Pump Efficiency	45%
Grid CO₂	0.71 kg/kWh
On-Site Use Vs Dhundi	72% vs 35%

A7 Capex And Finance

PV, Pump, Drive	₹9.50 L
Drip ₹1.12 L/ha × 8	₹8.96 L
Cold Room, 5 t	₹12.00 L
Borewell, Tank; Sensors	₹2.5 L; ₹1.4 L
Dryer, Crates; Scale	₹2.5 L; ₹0.3 L
Gross / Net Capex	₹37.16 L / ₹28.06 L
AIF 6% For 7 Yrs: Loan	₹25.25 L
Instalment; Per Farmer	₹4.52 L; ₹30,117

A4 Water, Yield, Inputs

Water Saved, Base	35%
Yield Gain, Base	+20%
Yield Gain, Conservative	+10%
Drip Extra Head (A)	10 m
Fertiliser Saved	11% of ₹20k/ha
Labour Saved	30% of ₹15k/ha
Bulk Buying, Not Booked	10%
Pump Energy Saved By Drip	9%

A8 Variants

Diesel Saved, V2 Hub	₹4,53,600
V2 Payback	1.9 yrs
AWD Water / Methane Cut	30% / 48%
Membrane, V3	₹3.0 to 3.9 L/ha
Exchange Rate	₹87 per \$
Hill Subsidy, V4	50%
Agrivoltaic Yield Floor	80%
Diesel Hub CO₂ Saved	13.5 t

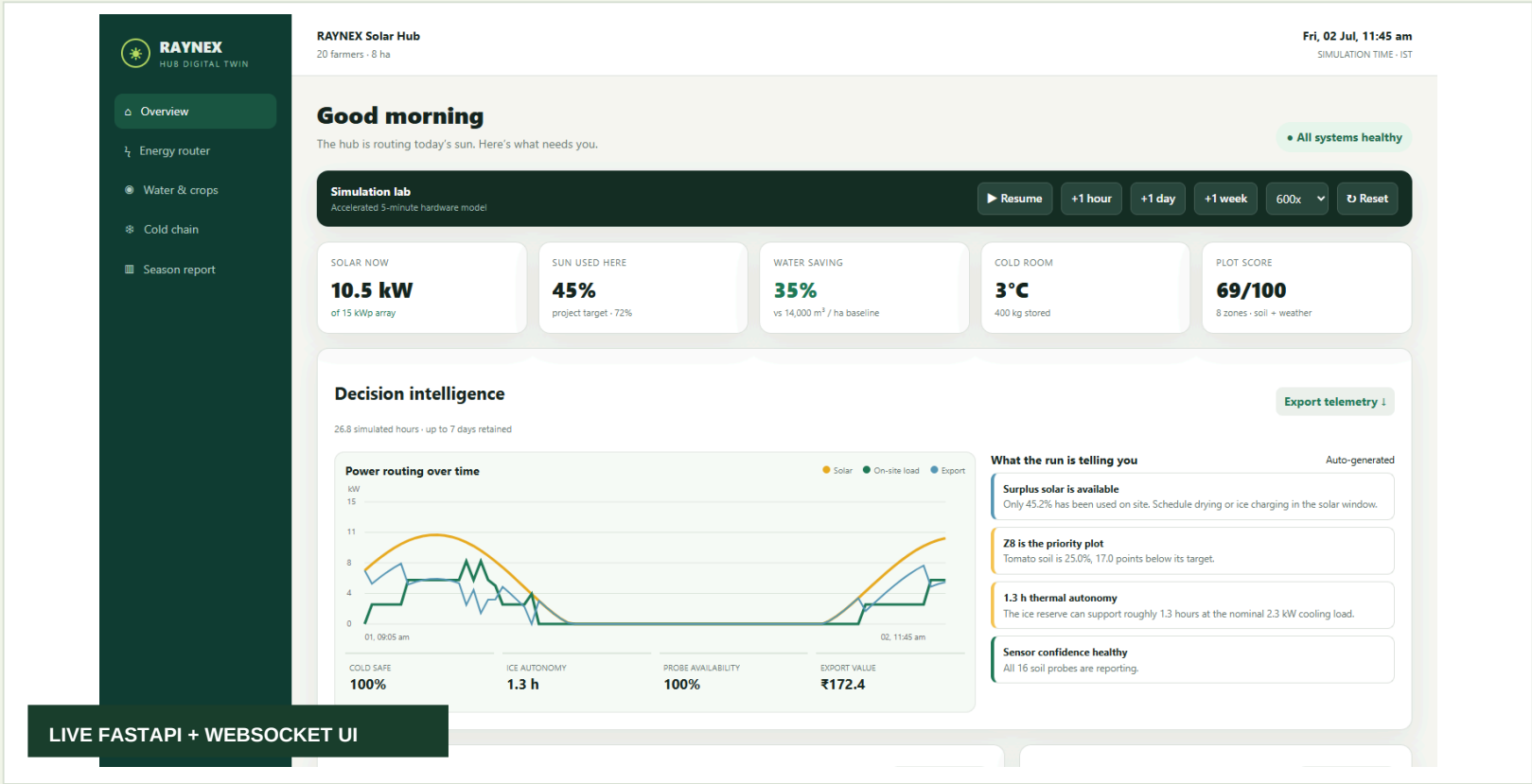
A11 Sensitivity: Payback In Years, Yield Gain × Cold-Room Cost

Yield	₹8 L	₹12 L	₹15.1 L	✔ Every cell pays back inside the 7-year AIF loan
10%	3.5	3.9	4.2	⚠ Weakest case, 10% yield and a ₹15.1 L room: 4.2 years
20%	2.3	2.5	2.7	🕒 No PV subsidy: 2.5 → 2.7 yrs; no bonus: 2.6 yrs
30%	1.7	1.9	2.1	₹ Conservative stack: 6.4 yrs grid, 3.2 diesel; farmer at break-even (–₹693)

Accelerate time, stress the system and turn every control cycle into evidence.

Not just live status: the lab explains what happened, why it matters and what to do next.

D4: Prototype · E5: Technical Approach



RUN THE SYSTEM

- Time - 5-min physics; +1 hour, day or week
- Speed - 1x, 60x, 600x or 3000x
- Stress - weather, outage, probe and door faults

EXPLAIN THE OUTCOME

- Trends - solar, on-site load and export
- Insights - surplus, driest plot, sensor confidence
- Resilience - cold safety and ice autonomy

VALIDATE THE CASE

- Claims - 72% on site, 35% water, cold <= 7 C
- Ledger - exact energy balance + event history
- Evidence - CSV telemetry and JSON run report

D4 SOFTWARE
PROTOTYPE

Before hardware

1 ACCELERATE

Run an hour, day or week in seconds

2 STRESS

Inject weather, grid and device faults

3 DIAGNOSE

Read trends and automatic findings

4 EVIDENCE

Compare targets; export the run